

Term paper on reproducing “Making Monolingual Sentence Embeddings Multilingual using Knowledge Distillation”

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Abstract

In this work, we replicate the paper (Reimers and Gurevych, 2020) by testing how well the method of distilling knowledge from an English-speaking teacher to a multilingual student performs in a resource-constrained setting. We use the compact SBERT MiniLM model as a frozen teacher and XLM-RoBERTa as a student, which is trained on English–German parallel data from the OPUS corpus by minimizing the MSE loss between student and teacher embeddings. Two types of tasks are evaluated: semantic text similarity (STS-2017), which requires predicting the degree of similarity between sentences, and Tatoeba, which requires the model to find the correct translation among a set of candidates. The results demonstrate significant improvement after distillation: Spearman’s ρ increases from 52.2 to 73.9 on EN–EN monolingual track and from 21.3 to 63.8 on EN–DE multilingual track, while the retrieval accuracy in Tatoeba increases from 41.5% to 83.1%. Although the absolute values are lower than in the original study, the overall dynamics is fully reproduced. This confirms the robustness and reproducibility of the cross-lingual distillation method, even with limited data and a more lightweight teacher. The project is available at [our Github repository](#).

1 Introduction

Sentence embeddings play a key role in a variety of natural language processing tasks: semantic textual similarity (STS) (Cer et al., 2017), information retrieval (Reimers and Gurevych, 2019; Thakur et al., 2021), clustering, and especially bitext mining and multilingual retrieval (Artetxe and Schwenk, 2019; Feng et al., 2022). An important challenge is to create embeddings that maintain high quality not only within a single language but also in multilingual scenarios (Feng et al., 2022).

In (Reimers and Gurevych, 2020), an approach based on the knowledge distillation method

(Hinton et al., 2015) was proposed. In its classical form, knowledge distillation involves transferring information from a more complex and accurate model (the teacher model) to a more compact or specialized model (the student model). This is usually achieved by minimizing the differences between the teacher and student outputs. In the current study, distillation is applied not to class probabilities, but to vector representations: the student model is trained to reproduce the embeddings of the teacher model for parallel sentences. The main reason for developing the knowledge distillation method was related to the fact that existing models individually have limitations: mBERT works well across languages but poorly at the sentence level, while SBERT produces high-quality sentence embeddings but only for English.

The authors of (Reimers and Gurevych, 2020) used the English-language Sentence-BERT (SBERT) model (Reimers and Gurevych, 2019) as a teacher with the multilingual XLM-RoBERTa model (Conneau et al., 2020) as one of the student models. The training consisted of minimizing the mean squared error (MSE) between the embeddings of the two models on parallel data. Experiments showed that this approach provides high quality on both monolingual and cross-lingual STS tasks and demonstrates the ability for zero-shot transfer (Reimers and Gurevych, 2020).

Replication of such results is useful to test their robustness and reproducibility. Especially, whether the method is applicable under resource constraints that researchers might face. Besides, it allows us to ensure that the results from the paper are still up-to-date, considering that the teacher model the authors used is being deprecated and is no longer recommended for usage in the community.

In this paper, we perform a targeted replication of the key experiments of the original paper, fo-

cusing on the English–German (EN–DE) pair. In particular, we:

- use SBERT as a teacher and XLM-RoBERTa as a student;
- prepare parallel EN↔DE data from the OPUS corpus; (Tiedemann, 2012);
- train the student model using MSE loss;
- conduct evaluation on STS 2017 (monolingual: EN–EN; cross-lingual: EN–DE);
- test quality on Tatoeba pair mining task.

Unlike the original paper, we excluded the BUCC bitext mining task as it requires large-scale resources and complex thresholding. This choice allows us to focus on the main hypothesis: is it possible to effectively transfer knowledge from an English-speaking teacher to a multilingual student using knowledge distillation?

2 Method

2.1 Models

We use a teacher-student scheme, where both models are specified via configuration and can be easily replaced. In our experiments, the default teacher is *all-MiniLM-L6-v2* from the Sentence-Transformers library, and the default student is *xlm-roberta-base*. The original paper used the *bert-base-nli-stsb-mean-tokens* (SBERT) teacher model, trained on English data (NLI + STS). However, this model is currently deprecated in the Sentence-Transformers repository and is not recommended for use as it produces sentence embeddings of low quality. Therefore, this teacher model was not used in our replication.

The teacher serves as a source of fixed English embeddings and remains unchanged during training. The student is based on the XLM-R architecture, and sentence embeddings are formed using mean pooling on top of hidden token states weighted by an attention mask. Since the dimensions of teacher and student differ, a trainable linear projection layer is added to the student, which aligns the vector dimensions.

2.2 Training data

We use the parallel Europarl EN–DE corpus from the OPUS collection as training data. The dataset is downloaded in streaming mode. A shortened

version of 1,000 sentences (subset) is used for debugging, and the first 50,000 sentences (full) are used for the main training. Before training, sentence pairs undergo minimal preprocessing: extra spaces are removed and repeated spaces within the text are collapsed; sentences shorter than two characters are excluded. The data is then randomly split into training (90%) and validation (10%) sets.

2.3 Training objective

Following the original paper (Reimers and Gurevych, 2020), student learning is formulated as minimizing the mean square error between the teacher’s embeddings for the English sentence and the student’s embeddings for both English and German:

$$\frac{1}{|B|} \sum_{j \in B} \left[(M(s_j) - \hat{M}(s_j))^2 + (M(s_j) - \hat{M}(t_j))^2 \right],$$

where M — teacher-model, $\hat{M}$ — student, s_j — English sentence, t_j — German translation, and B — current batch.

In our implementation, this function corresponds to the sum of two MSE terms:

- $\text{MSE}(\text{student}(s_j), \text{teacher}(s_j))$
- $\text{MSE}(\text{student}(t_j), \text{teacher}(s_j))$

Before calculating the error, all vectors are L2-normalized. This is how the student adapts both its own representation of the English sentence and the representation of the German translation to the teacher’s vector for English.

2.4 Training procedure

Training was performed using the AdamW optimizer. The batch size, number of epochs and learning rate are specified in the configuration file and were taken from the original paper for reproducibility (batch size=64, learning rate=2e-5, epochs=20). After each epoch, the model was evaluated on the 10% validation set, where the mean squared error and cosine similarity between the student vectors and the teacher’s reference vectors were calculated. The [Weights & Biases](#) platform was used to monitor metrics and training dynamics.

2.5 STS evaluation

To assess the quality of sentence embeddings, we used the STS-2017 cross-lingual dataset available in the MTEB library. Two tracks were considered:

monolingual EN–EN and cross-lingual EN–DE. For each track, sentences were encoded by the selected model (teacher or student), the embeddings were normalized and the cosine similarity was calculated for each pair. Quality was assessed using Spearman correlation (values are reported as $\rho \times 100$) between the predicted similarities and human ratings.

2.6 Tatoeba evaluation

Additionally we evaluated the quality using the Tatoeba EN–DE dataset (Helsinki-NLP/tatoeba_mt). The first 1,000 test sentence pairs were selected for the experiment, as in the original paper. Each English sentence was encoded using the selected model and the closest German candidate was checked for match with the correct translation based on cosine similarity. A similar procedure was performed for the DE→EN direction. The metric was the top-1 accuracy, averaged over both directions: we measure how often the correct answer is ranked first in each direction and then take the average of those scores.

3 Results

The original paper (Reimers and Gurevych, 2020) demonstrated that student training through knowledge distillation successfully transfers the qualities of an English teacher to a multilingual student. The resulting models achieved high correlations on STS-2017 (e.g., EN–EN $\approx$ 79, EN–DE $\approx$ 75 Spearman ρ) and demonstrated strong results on bitext mining tasks (BUCC, Tatoeba), often outperforming larger multilingual models without distillation.

The quantitative results are summarized in Tables 1–3. Table 1 presents the performance on the monolingual EN–EN STS 2017 track, where distillation boosts correlation from 52.2 to 73.9. Table 2 shows the cross-lingual EN–DE track with correlation rising sharply from 21.3 to 63.8. Finally, Table 3 reports results on the Tatoeba retrieval task, where accuracy improves from only 41.5% before distillation to 83.1% after training, in both EN→DE and DE→EN directions.

As mentioned above, the learning process was monitored using [Weights & Biases](#). The resulting curves presented in the Section A show that MSE for training and validation gradually decreases,

while the cosine similarity between the teacher and student embeddings increases. This demonstrates that the student is successfully approaching the quality of the teacher’s embeddings. The code of this project is available at [our Github repository](#).

Model	EN–EN
XLM-R (student, before distillation)	52.2
XLM-R (student, after distillation, 5 epochs)	67.9
XLM-R (student, after distillation, 20 epochs)	73.9

Table 1: Performance on STS 2017 monolingual track (Spearman’s $\rho \times 100$).

Model	EN–DE
XLM-R (student, before distillation)	21.3
XLM-R (student, after distillation, 5 epochs)	57.8
XLM-R (student, after distillation, 20 epochs)	63.8

Table 2: Performance on STS 2017 cross-lingual track (Spearman’s $\rho \times 100$).

Direction	Model (XLM-R)	
	Before distillation	After distillation
EN→DE @1 (%)	37.7	83.1
DE→EN @1 (%)	45.3	83.2
Avg. (%)	41.5	83.1

Table 3: Tatoeba EN–DE: accuracy@1 for each direction and the average, before and after distillation.

4 Discussion

The results of the replication show that knowledge distillation effectively transfers semantic properties from an English-speaking teacher to a multilingual student. Before training, XLM-R shows low results in both monolingual and, especially, cross-lingual modes, but after distillation these results increase significantly. This improvement is particularly noticeable on the Tatoeba translation task: after distillation, the model reaches approximately 83% in both directions. Considering the obtained results, we can confirm that distillation improves the processing within English and makes the model a reliable tool for cross-lingual sentence matching.

Compared to the original work by (Reimers and Gurevych, 2020) our absolute values are slightly lower, which could be explained by two factors: the use of a simpler teacher and a smaller amount of parallel data (only 50k). Nevertheless,

the overall dynamics is fully reproduced: in all cases we observed a sharp increase in quality after distillation. Thus, it demonstrates the robustness and reproducibility of the method even with limited resources.

Despite only one multilingual track used in this replication, the method can also be particularly useful for different low-resource languages where there is no annotated data and only parallel sentences can be used.

5 Conclusion

In summary, the present study clearly demonstrated the viability of the knowledge distillation method under resource constraints. Despite using a more compact teacher and a smaller dataset than in the original study, we still observed the overall improvement pattern: the student gradually brought their vectors closer to the teacher’s embeddings, and the test task results confirmed a significant improvement in quality. This was particularly noticeable in cross-lingual scenarios, where the model initially demonstrated extremely low performance but, after training, achieved high values. Distillation thus solves a key problem: multilingual models like mBERT and XLM-R cover dozens of languages but are unable to generate reliable sentence embeddings, while English-language SBERT models produce high-quality vectors, but only for a single language. The distillation method combines the strengths of both approaches and enables the construction of multilingual models for searching and matching meanings even when large, annotated corpora are missing. Our study not only confirms the findings of the original work but also demonstrates that their approach is successfully replicated even under simplified conditions, confirming the practical value of the method.

6 Limitations

While our replication successfully demonstrates the viability of knowledge distillation for multilingual sentence embeddings under resource constraints, there are several limitations to consider:

- **Dataset size and diversity:** We only used a subset of 50,000 parallel EN–DE sentences from the OPUS corpus. The results might differ with larger or more diverse datasets.
- **Language coverage:** Our study focuses

solely on English–German. The method’s performance on other language pairs, particularly low-resource languages, has not been evaluated in this work, though we see great potential with them.

- **Teacher model choice:** We used a smaller teacher (all-MiniLM-L6-v2) due to availability and resource constraints. A larger teacher might yield different results and higher embedding quality.
- **Evaluation tasks:** We excluded the BUCC bitext mining task and only evaluated on STS-2017 and Tatoeba. Therefore, the generalizability of results to other tasks remains untested.

These limitations suggest that while the replication is successful in demonstrating the method’s robustness under constrained conditions, caution should be taken when generalizing the results to larger datasets, other languages or more complex tasks.

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A Appendix

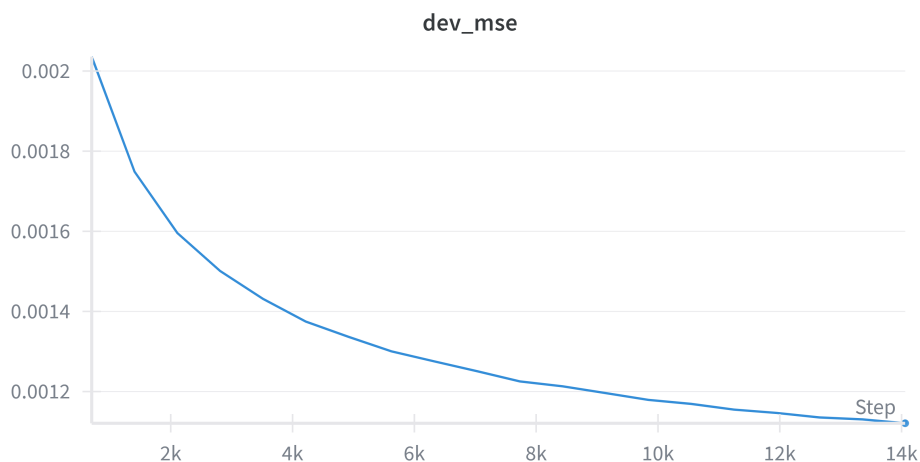


Figure 1: MSE measured on the development set.

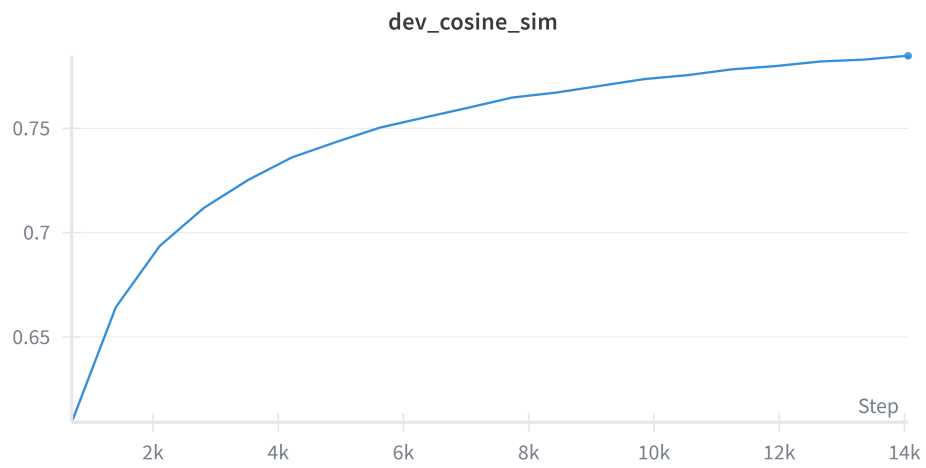


Figure 2: Cosine similarity measured on the development set.

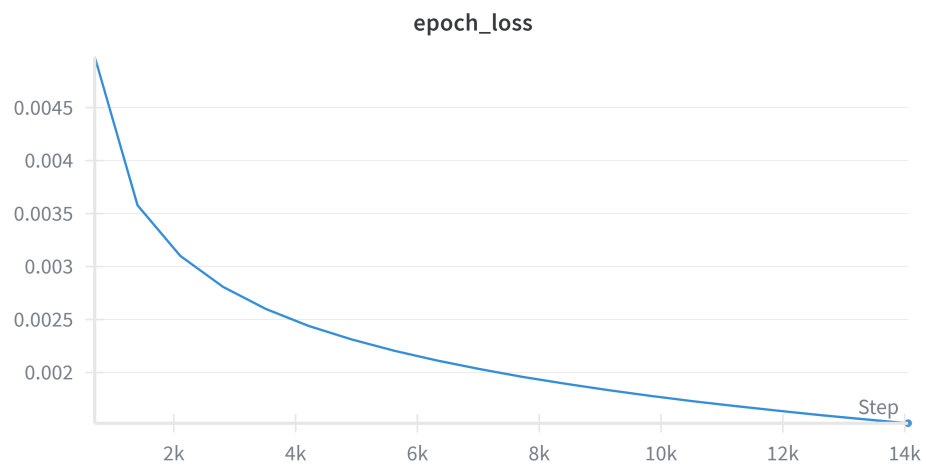


Figure 3: Training loss dynamics per epoch.